WAS Experiment 01

Aim: Install Burp Suite in Kali Linux

Learning Objective: To demonstrate the installation and basic configuration of Burp Suite in Kali Linux and to understand how Burp Suite can be used as an intercepting web proxy for analyzing HTTP/HTTPS traffic between a web browser and a web application.

Learning Objectives:
After completing this experiment, students will be able to:

- Understand the concept of Burp Suite.
- Understand the role of an intercepting proxy in web application security.
- Install Burp Suite in Kali Linux.
- Launch Burp Suite and configure a temporary project.
- Understand the Proxy functionality of Burp Suite.
- Configure a web browser to work with Burp Suite.
- Intercept HTTP requests and responses.
- Analyse web application traffic.
- Understand the purpose of Burp Suite tools such as Proxy, Repeater, Intruder, and Scanner.
- Apply Burp Suite in a controlled web security testing environment.

Tools: Kali Linux, Burp Suite Community Edition, Web Browser - Firefox, Terminal and Internet connection.

Theory:

Introduction

Web applications communicate with servers using HTTP and HTTPS protocols. During normal browsing, requests and responses are exchanged directly between the web browser and the web server. Security professionals need to inspect this communication to understand how web applications process user input, authentication requests, cookies, headers, parameters, and other HTTP information.

Burp Suite is an integrated web application security testing platform developed by PortSwigger. It provides tools that allow security testers to intercept, inspect, modify, and replay HTTP/HTTPS requests and responses.

Burp Suite is commonly used during web application penetration testing and vulnerability assessment.

What is Burp Suite?

Burp Suite is a collection of tools used for testing the security of web applications.

It works primarily as an intercepting proxy between a web browser and a web server.

The basic communication flow can be represented as:

Browser → Burp Suite → Web Server

When Burp Suite is configured as a proxy, HTTP/HTTPS requests from the browser can be captured by Burp Suite before being forwarded to the web server.

Similarly, responses from the web server can be inspected before being returned to the browser.

What is an Intercepting Proxy?

An intercepting proxy is a security testing component that sits between the client and the server.
It allows the tester to:

- Capture HTTP requests.
- View request headers.
- View request parameters.
- Modify requests before forwarding them.
- Inspect server responses.
- Analyse cookies and session information.
- Replay requests.
- Study application behaviour.

Major Components of Burp Suite

1. Proxy

The Proxy tool intercepts HTTP and HTTPS traffic between the browser and the target web application.
It allows the tester to:

- Intercept requests.
- Forward requests.
- Drop requests.
- Modify requests.
- Inspect responses.

2. Repeater

Repeater allows a captured HTTP request to be sent repeatedly.
A tester can modify parameters, headers, or other parts of a request and observe how the server responds.

3. Intruder

Intruder is used for automated manipulation of request parameters.
It can be useful for authorized security testing such as:

- Parameter testing.
- Fuzzing.
- Input validation testing.
- Controlled credential testing.

4. Scanner

Burp Suite's automated scanning functionality can help identify web application security vulnerabilities.
Availability and functionality depend on the Burp Suite edition.

5. Decoder

Decoder can encode or decode data in different formats, helping testers analyse encoded values encountered during testing.

6. Comparer

Comparer allows two pieces of data, such as HTTP requests or responses, to be compared.

Procedure:

Step 1: Start Kali Linux

Start the Kali Linux operating system.

Open the Kali Linux desktop and launch the Terminal.

Expected Output:

Kali Linux starts successfully and the desktop environment is displayed.

Step 2: Update the Package Repository

Open the Terminal and execute:

sudo apt update

Enter the password when prompted.

This updates the local package information available to the Kali Linux package manager.

Expected Output:

The package repository information is successfully updated.

Step 3: Install Burp Suite

Execute the following command:

sudo apt install burpsuite

If prompted to confirm the installation, enter:

Y

The required Burp Suite package and dependencies are installed.

Expected Output:

The terminal displays the installation progress and completes successfully.

Step 4: Launch Burp Suite

After installation, Burp Suite can be launched from the Kali Linux application menu.

Navigate to:

Applications → Web Application Analysis → Burp Suite

Alternatively, open the Terminal and execute:

burpsuite

Expected Output:

The Burp Suite startup window appears.

Step 5: Select the Project Type

When Burp Suite starts, select:

Temporary project

Then continue with the default configuration.

Expected Output:

The Burp Suite main interface opens.

Step 6: Open the Proxy Tab

In the Burp Suite interface, select:

Proxy → Intercept

The Proxy section provides functionality for intercepting HTTP requests.

Expected Output:

The Proxy/Intercept interface is displayed.

Step 7: Configure the Browser Proxy

Open Firefox in Kali Linux.

Configure the browser to use Burp Suite as its proxy.

Use:

Proxy Address: 127.0.0.1

Proxy Port: 8080

The address 127.0.0.1 refers to the local Kali Linux system, while port 8080 is commonly used by Burp Suite's proxy listener.

Expected Output:

Firefox is configured to send its web traffic through Burp Suite.

Step 8: Verify the Burp Proxy Listener

In Burp Suite, navigate to:

Proxy → Proxy settings

Check the proxy listener configuration.

The listener should be configured for the local address and the appropriate proxy port.

Typical configuration:

Interface: 127.0.0.1

Port: 8080

Expected Output:

The Burp Suite proxy listener is running successfully.

Step 9: Enable Intercept

Go to:

Proxy → Intercept

Ensure that interception is enabled.

The interface should indicate that Burp Suite is ready to intercept requests.

Expected Output:

Burp Suite waits for HTTP requests from the configured browser.

Step 10: Generate Web Traffic

In Firefox, access a web page or a web application that you are authorized to test.

For example, a local test application can be used:

http://localhost/

When the browser sends a request, Burp Suite captures it.

Expected Output:

The HTTP request appears inside the Burp Suite Proxy/Intercept window.

Step 11: Analyse the Intercepted Request

Inspect the captured request.

A typical HTTP request may contain:

GET / HTTP/1.1

Host: localhost

User-Agent: Mozilla/5.0

Accept: text/html

The request may contain several components such as:

- HTTP method
- URL/path
- HTTP version
- Host header
- User-Agent
- Cookies
- Parameters
- Other HTTP headers

Expected Output:

The HTTP request and its headers are displayed in Burp Suite.

Step 12: Forward the Request

After examining the request, click:

Forward

Burp Suite forwards the intercepted request to the web server.

Expected Output:

The request is forwarded and the corresponding server response is received.

Step 13: Inspect the HTTP Response

The response returned by the web server can be analysed in Burp Suite.

A response may contain:

HTTP/1.1 200 OK

Content-Type: text/html

The response may also contain:

- HTTP status code
- Response headers
- Cookies
- HTML content
- Server information
- Other response data

Expected Output:

The HTTP response is successfully displayed and can be analysed.

Step 14: Turn Interception Off

After testing, return to:

Proxy → Intercept

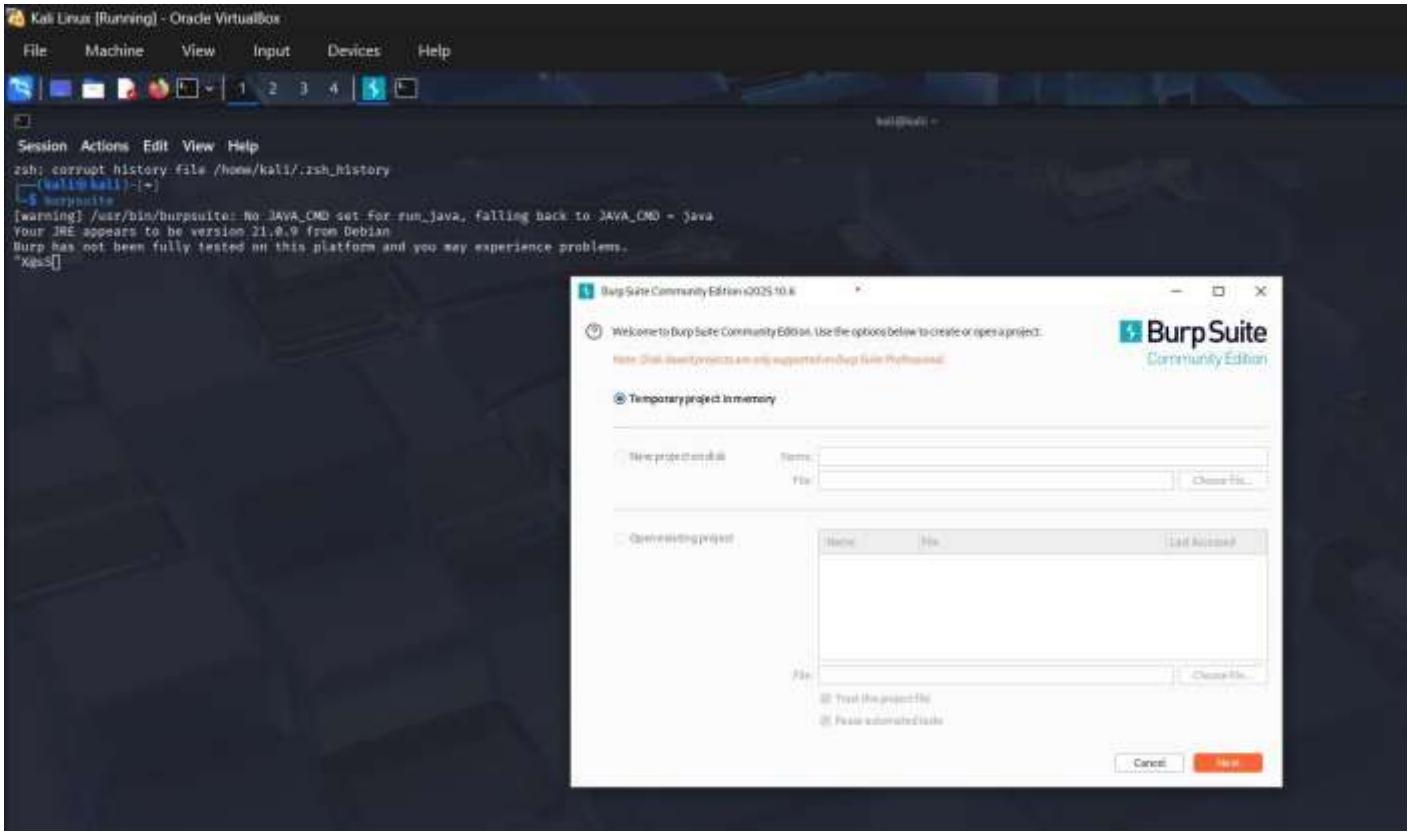
Disable interception.

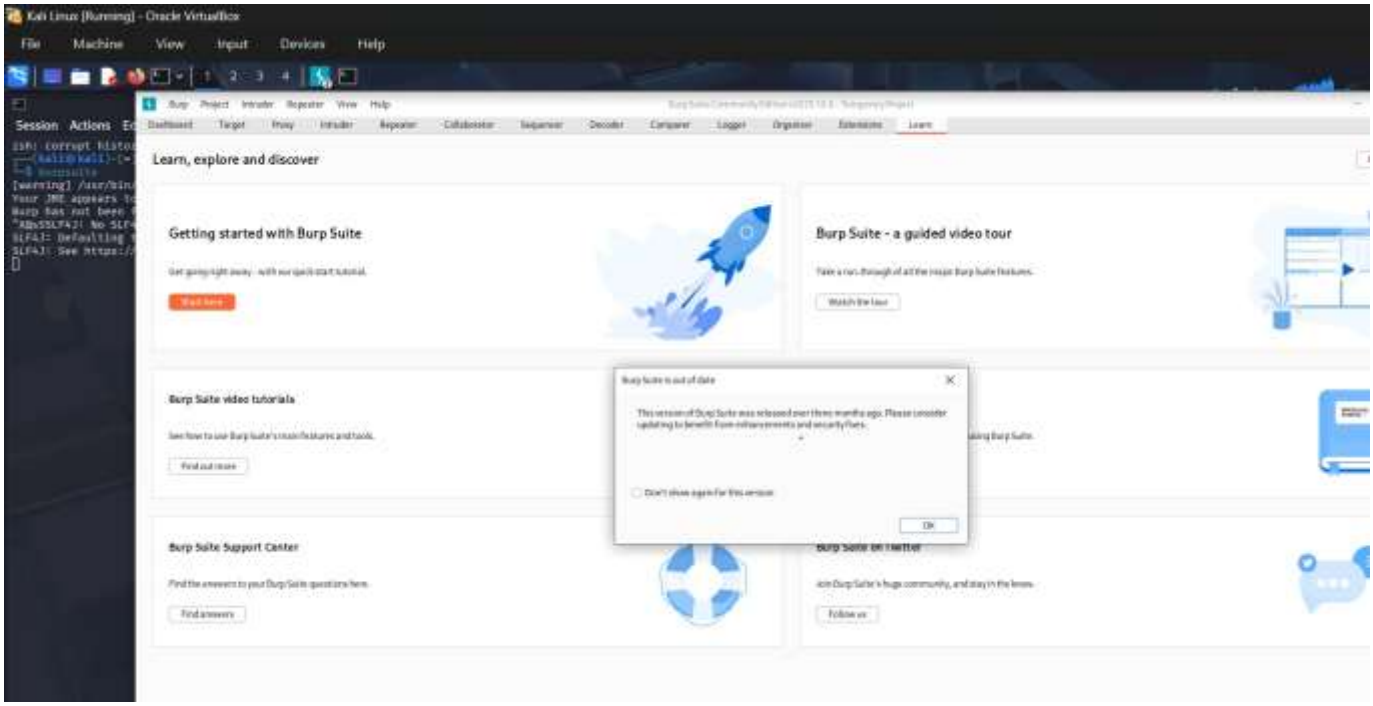
This allows browser requests to pass through the proxy without being manually stopped.

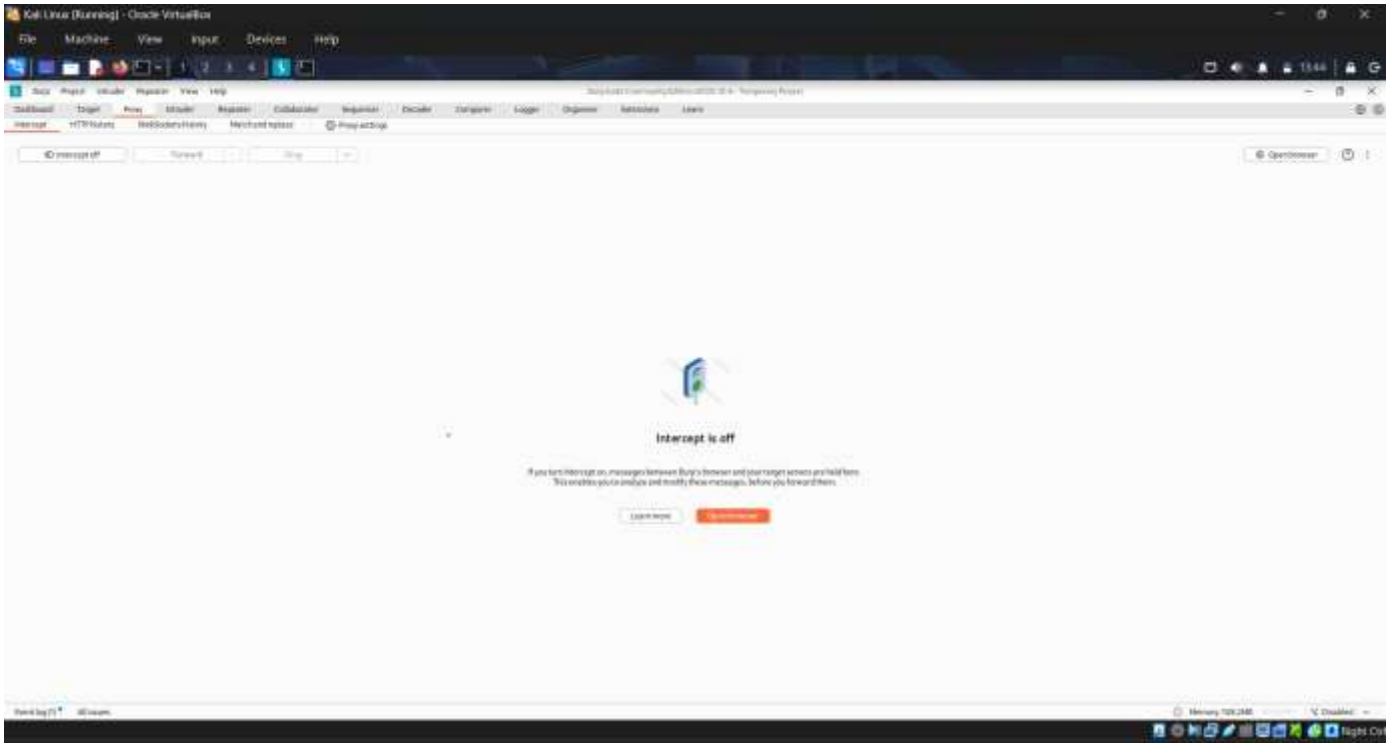
Expected Output:

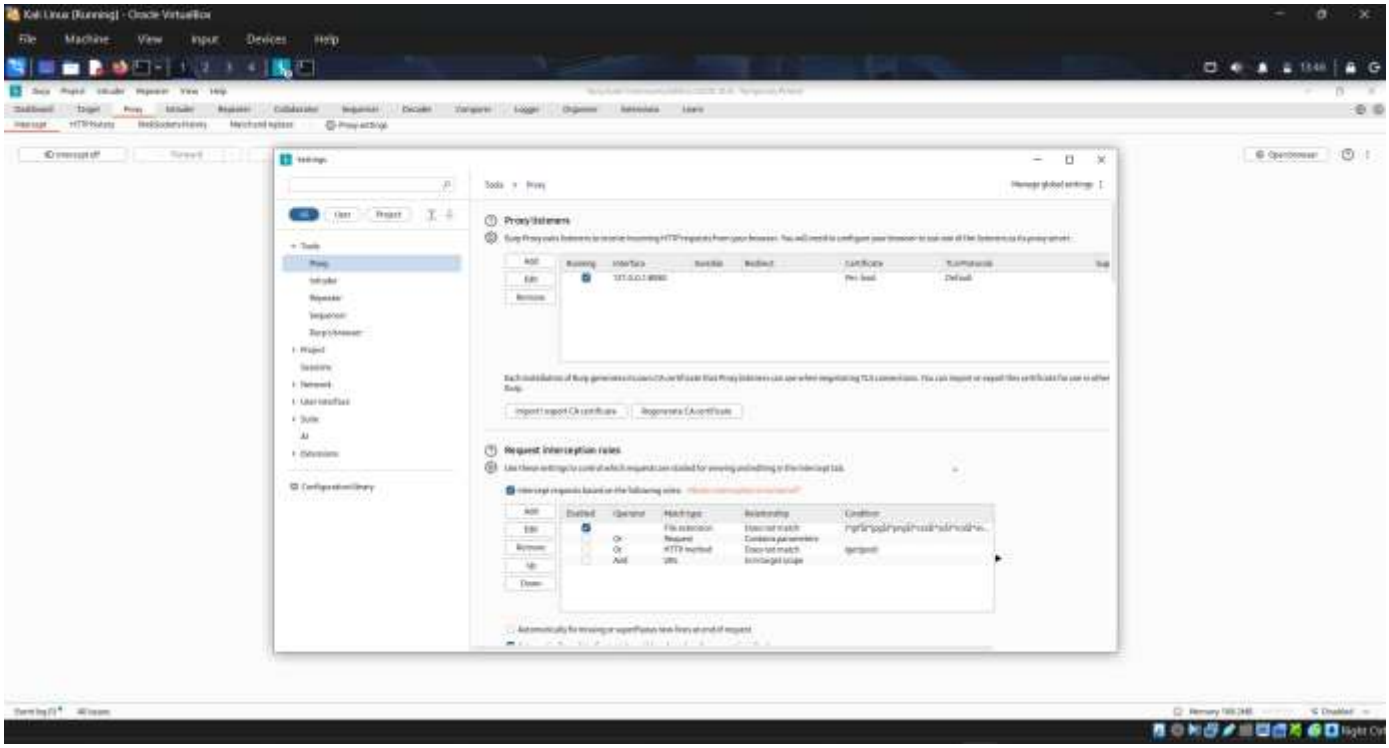
Browser traffic passes through Burp Suite without requiring manual forwarding.

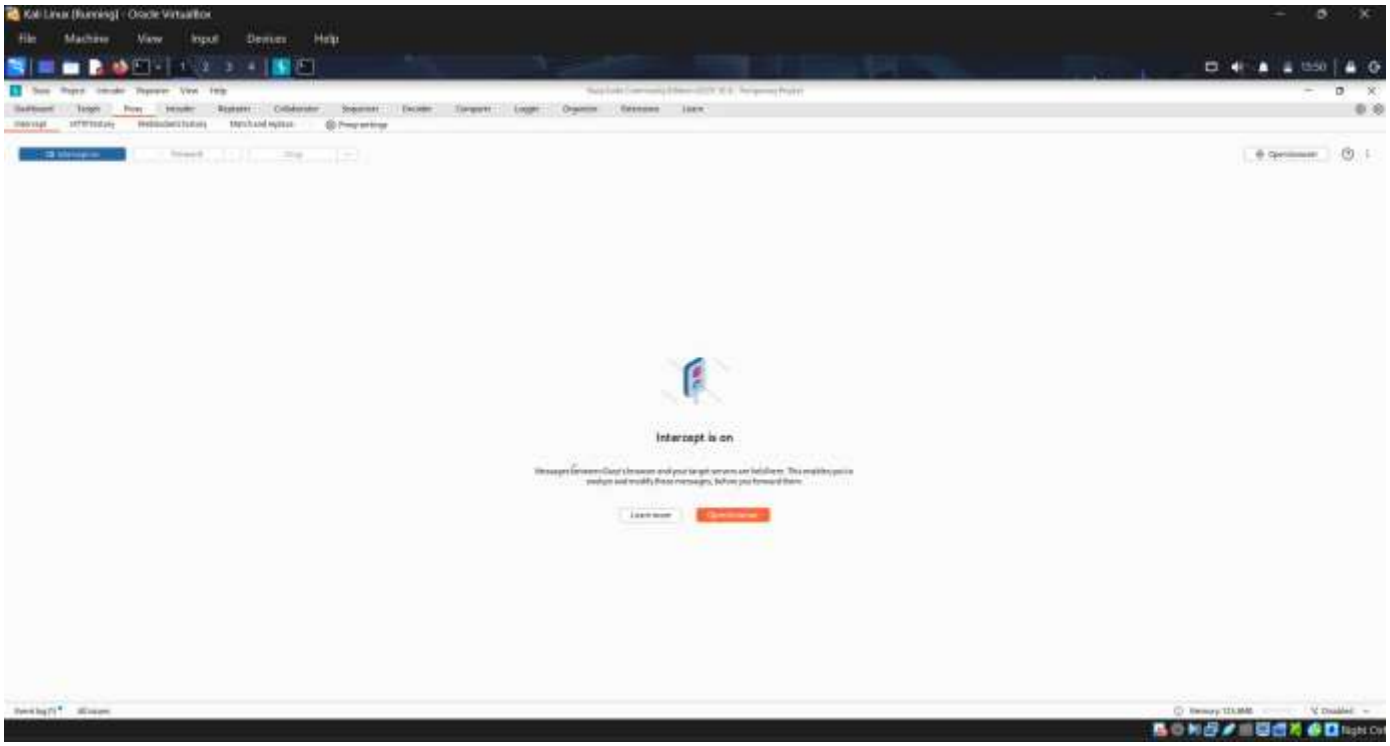
Outputs:

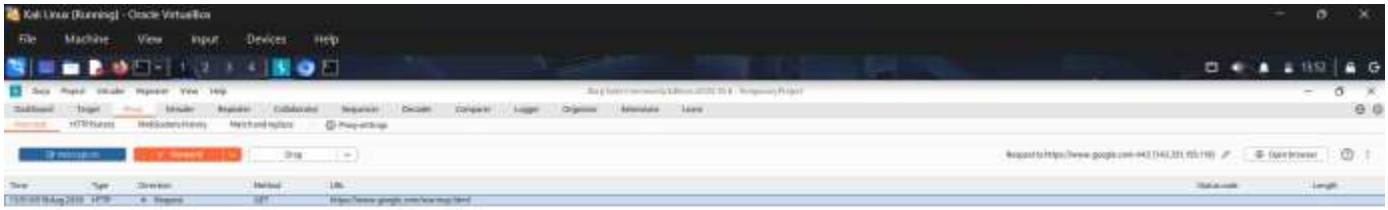

Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
[warning] /usr/bin/burpsuite: No JAVA_CMD set for run_java, falling back to JAVA_CMD = java
Your JRE appears to be version 21.0.9 from Debian
Burp has not been fully tested on this platform and you may experience problems.
Burp Suite Community Edition
Welcome to Burp Suite Community Edition. Use the options below to create or open a project.
Temporary project in memory
Cancel


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Learn, explore and discover
Getting started with Burp Suite
Burp Suite - a guided video tour
Burp Suite video tutorials
Burp Suite Support Center
OK


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Intercept is off


Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Proxy listeners
Request interception rules







Observation:

The following observations are made during the experiment:

1. Burp Suite was successfully installed in Kali Linux.
2. Burp Suite was successfully launched.

3. The browser was configured to use Burp Suite as a proxy.
4. Burp Suite successfully intercepted web requests.
5. HTTP request headers and parameters could be inspected.
6. Intercepted requests could be forwarded to the web server.
7. Server responses could be inspected through Burp Suite.
8. Burp Suite acts as an intermediary between the browser and the web server.

Result and Discussion:

The experiment successfully demonstrated the installation and basic configuration of Burp Suite in Kali Linux.

Burp Suite was configured as an intercepting proxy between the web browser and the web server. HTTP requests generated by the browser were intercepted and analysed using the Proxy module.

The experiment demonstrated that Burp Suite can provide visibility into web application communication, including HTTP methods, headers, parameters, cookies, and server responses.

The use of Burp Suite is particularly useful in web application security testing because it allows security professionals to understand how applications process and respond to different HTTP requests.

All testing should be performed only against systems and web applications for which authorization has been obtained.

Conclusion:

This experiment demonstrated the installation and basic operation of Burp Suite in Kali Linux. Burp Suite was configured as an intercepting proxy and successfully used to capture and analyse web traffic between a browser and a web server.

The experiment provided an understanding of the role of intercepting proxies in web application security testing and introduced important Burp Suite components such as Proxy, Repeater, Intruder, Scanner, Decoder, and Comparer.

Burp Suite provides an effective environment for analysing web application traffic and performing authorized security assessments.

Viva Questions:

1. What is Burp Suite?
2. What is an intercepting proxy?
3. What is the purpose of the Proxy module in Burp Suite?

For Faculty Use:

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				